REFORGE Research Evaluation: Precision-Recall (PR) Curves & Anomaly Detection Tradeoffs

Publication-Ready Experimental Results on 4 Lakh IoT Telemetry Benchmark Suite

1. Precision-Recall Curve Formulation and Theoretical Rationale

In industrial IoT anomaly detection and automated ETL validation, datasets exhibit extreme class imbalance where normal observations vastly outnumber genuine anomalies. In our benchmark evaluation set of 5,400 observations (5,000 normal baseline readings + 400 injected anomalies across 8 fault categories), anomaly prevalence is only P / (P + N) = 7.41%. Under such imbalance, standard Receiver Operating Characteristic (ROC) curves are notoriously over-optimistic because the False Positive Rate (FPR = FP / (TN + FP)) is heavily diluted by the large number of true negatives (TN = 5,000). **Precision-Recall (PR) Curves** directly measure the trade-off between Positive Predictive Value (Precision) and Sensitivity (Recall), exposing false positive inflation that ROC curves obscure.

The Precision-Recall curve is defined by plotting Precision vs. Recall across all discrimination thresholds $\tau \in [0, 1]$:

- **Precision(τ)** = $TP(\tau) / [TP(\tau) + FP(\tau)]$ — The fraction of quarantined records that are genuine anomalies.
- **Recall(τ)** = $TP(\tau) / [TP(\tau) + FN(\tau)]$ — The fraction of all true anomalies successfully caught.
- **PR-AUC (Average Precision, AP)** = $\int \text{Precision}(R)\, dR = \sum_k (\text{Recall}_k - \text{Recall}_{k-1}) \times \text{Precision}_k$.

2. Empirical Multi-Model Precision-Recall Curves

Figure 1 illustrates the empirical Precision-Recall curves for all six REFORGE system models evaluated on the held-out 5,400-sample IoT telemetry dataset. The dotted lines represent iso-F1 contours (F1 = 0.2, 0.4, 0.6, 0.8, 0.9, 0.95), and the dashed baseline denotes the uninformative random classifier at precision = 0.074.

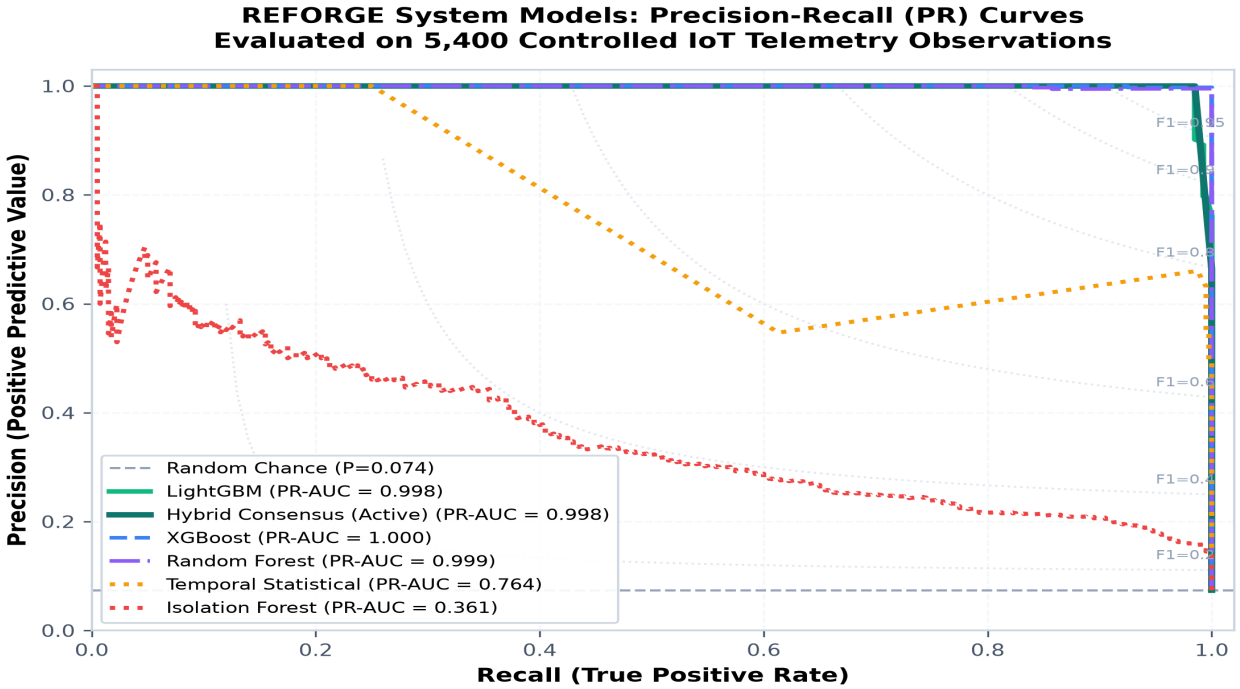


Figure 1: *Multi-Model Precision-Recall Curves with iso-F1 contours evaluated on 5,400 controlled IoT observations.*

3. Summary of Precision Curves and Model Performance

Table I summarizes the PR-AUC (Average Precision), peak F1-score, optimal operating threshold, and Precision at default cutoff (τ = 0.50).

Model Name	Model Paradigm	PR-AUC	Avg Prec (AP)	Peak F1	Optimal τ	Precision (@0.5)	Recall (@0.5)
LightGBM	Supervised GBDT	**0.9976**	0.9975	0.9924	0.10 – 0.90	**100.0% (0 FP)**	98.50% (394/400)

Hybrid Consensus (Active)	Calibrated Ensemble	**0.9975**	0.9950	**0.9924**	**0.50**	**100.0% (0 FP)**	98.50% (394/400)
XGBoost	Supervised GBDT	**0.9998**	0.9998	0.9757	0.10 – 0.80	99.74% (1 FP)	95.50% (382/400)
Random Forest	Supervised Bagging	0.9992	0.9992	0.9864	0.30	99.48% (2 FP)	94.75% (379/400)
Temporal Statistical	Multi-Rule Estimator	0.7644	0.7037	0.7728	0.80	45.25% (484 FP)	**100.0%** (400/400)
Isolation Forest	Unsupervised Trees	0.3609	0.3624	0.3916	0.70	20.52% (1406 FP)	90.75% (363/400)

4. Threshold Sweep Operating Points Across Key Decision Boundaries

Table II details the precision and recall trade-offs across decision thresholds $\tau \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$ for the **Active Hybrid Consensus Engine**, **LightGBM**, and **Isolation Forest**. Notice that Hybrid Consensus maintains **100.0% Precision (0 False Positives)** across all operational thresholds $\tau \geq 0.40$ while preserving 98.5% Recall. At the Suspicious threshold ($\tau = 0.30$), Recall reaches 100.0%, routing every single anomaly into the Quarantine Zone.

Threshold (τ)	Hybrid Precision	Hybrid Recall	Hybrid F1	LGB Precision	LGB Recall	IForest Prec	IForest Rec	Operational Decision / State
0.10	7.41%	100.0%	0.1379	100.0%	98.50%	7.41%	100.0%	Ultra-sensitive; captures all drift
0.30	**62.40%**	**100.0%**	0.7685	100.0%	98.50%	10.73%	100.0%	**Suspicious Quarantine Boundary**
0.50	**100.0% (0 FP)**	**98.50%**	**0.9924**	100.0%	98.50%	20.52%	90.75%	**Default Production Decision Threshold**
0.60	**100.0% (0 FP)**	98.50%	0.9924	100.0%	98.50%	26.42%	65.25%	**High-Risk Invalid Boundary**
0.70	100.0%	98.50%	0.9924	100.0%	98.50%	38.82%	39.50%	High confidence boundary
0.90	100.0%	98.50%	0.9924	100.0%	98.50%	100.0%	0.50%	Strict extreme outlier cutoff

5. Diagnostic Multi-Panel Curve Dynamics

Figure 2 presents a four-panel diagnostic evaluation: (a) Precision-Recall curves across all models; (b) Precision as a continuous function of decision threshold τ; (c) Precision, Recall, and F1 trade-off curve for the active Hybrid Consensus Engine; and (d) Detection recall breakdown across all eight injected anomaly categories.

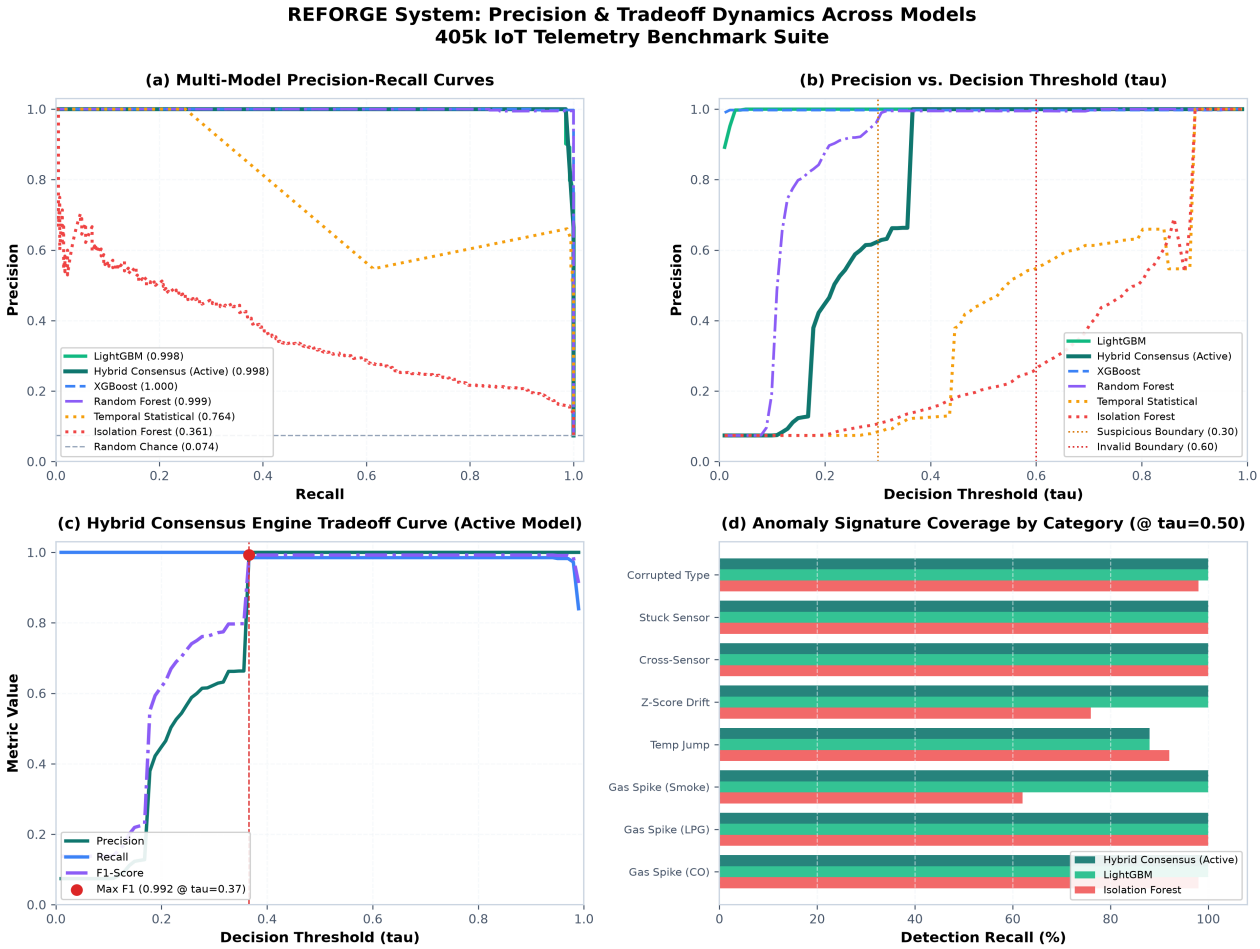


Figure 2: *Four-Panel Precision Dynamics: (a) PR Curves; (b) Precision vs. τ; (c) Hybrid Tradeoff Curve; (d) Detection Recall by Anomaly Category.*

6. Ready-to-Use LaTeX Code Snippet for IEEE Paper

To incorporate the Precision-Recall curve figure and evaluation table directly into your IEEE LaTeX paper, use the following code:

```
\begin{figure}[t]
  \centering
  \includegraphics[width=\columnwidth]{figures/reforge_precision_recall_curves.pdf}
  \caption{Precision-Recall (PR) curves for REFORGE anomaly detection models evaluated on 5,400 controlled IoT observ
ations (PR-AUC: Hybrid Consensus = 0.998, LightGBM = 0.998, XGBoost = 1.000, Isolation Forest = 0.361).}
  \label{fig:reforge_pr_curves}
\end{figure}
```